

Homework Submission — Lakebase-Powered Ticketing System

1. Databricks App URL

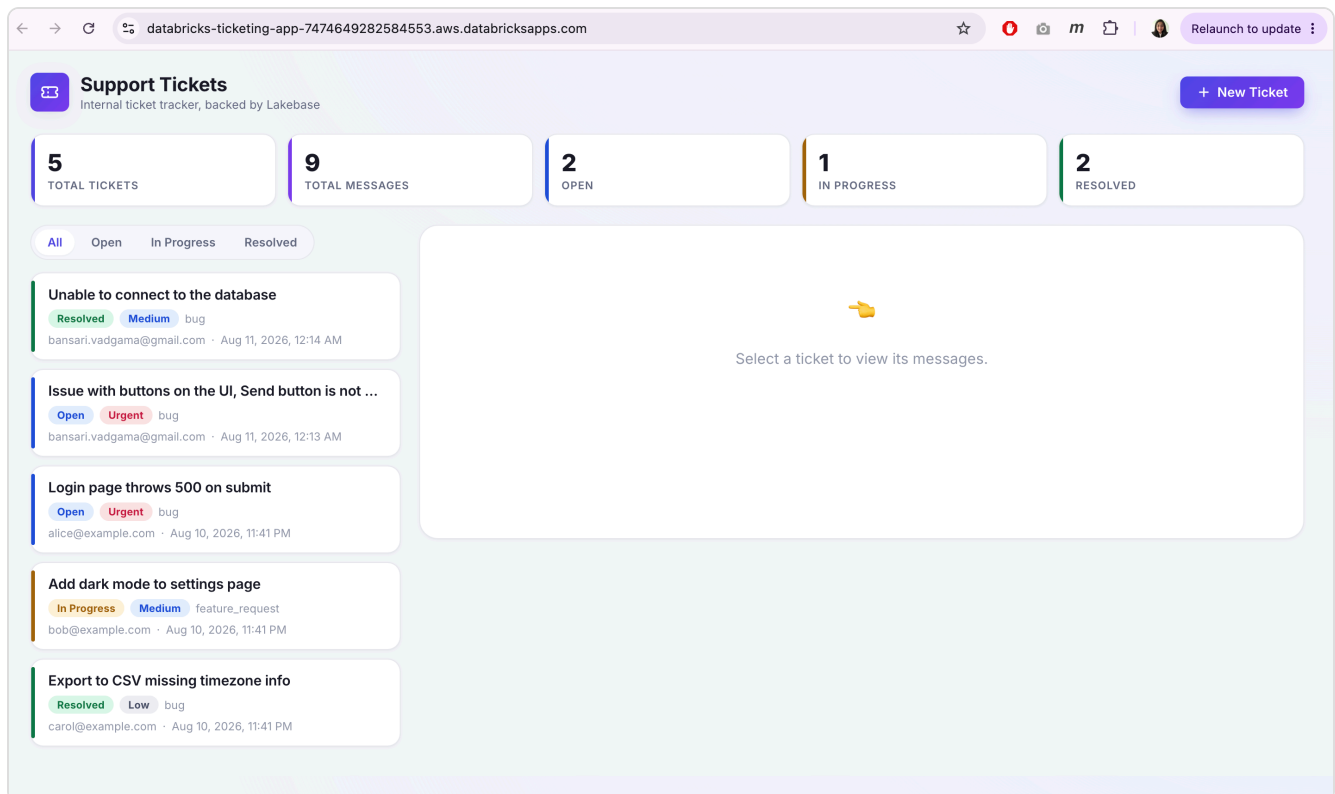
<https://databricks-ticketing-app-7474649282584553.aws.databricksapps.com>

2. Source code

Zipped from the project root (`ticketing-system/`) — see `ticketing-system-source.zip` submitted alongside this document. Contains:

- `sql/` — versioned schema (01–03) and the ownership fix applied during setup (04)
- `src/` — the Flask app (`app.py`, `lakebase.py`, `setup_secrets.py`, `app.yaml`, `templates/`, `static/`)
- `scripts/` — one-off local admin script used to recreate the schema under the app's own role
- `README.md` — full setup/deploy walkthrough

3. Screenshot of the deployed application



Stats strip and filter pills reflect live Lakebase data: 5 active tickets, 9 active messages, split across Open / In Progress / Resolved.

4. Screenshots of the Lakebase tables and sample records

tickets table — ticket_id, title, status, priority, category:

PROJECT

ticketing-syst...

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Tables

databricks_postgres

Schema

public

Search...

ticket_messages

tickets

ticket_id serial

title text

status text

priority text

category text

1

Login page throws 500 ...

open

urgent

bug

2

Add dark mode to setti...

in_progress

medium

feature_request

3

Export to CSV missing ...

resolved

low

bug

4

Issue with buttons on ...

open

urgent

bug

5

Unable to connect to t...

resolved

medium

bug

6

VPN not working , need...

resolved

high

bug

tickets table, scrolled to created_by / created_at / deleted_at — row 6 shows a populated deleted_at, from testing the delete-with-confirmation bonus feature (soft delete keeps the row instead of removing it):

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Tables

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Schema

public

Search...

ticket_messages

tickets

text

created_by text

created_at timestamp w...

deleted_at timestamp w...

ticket_messages

alice@example.com

2026-08-10 21:41:31.11...

NULL

ticket_messages

request

bob@example.com

2026-08-10 21:41:31.11...

NULL

ticket_messages

carol@example.com

2026-08-10 21:41:31.11...

NULL

ticket_messages

bansari.vadgama@gmail...

2026-08-10 22:13:24.39...

NULL

ticket_messages

bansari.vadgama@gmail...

2026-08-10 22:14:57.01...

NULL

ticket_messages

bansari.vadgama@gmail...

2026-08-10 22:16:35.15...

2026-08-10 22:17:09.98...

ticket_messages

ticket_messages table — message_id, ticket_id, message_text, author, created_at:

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Schema

public

Search...

ticket_messages

tickets

137ms

1 - 10 of 10

+

Add record

	message_id serial	ticket_id integer	message_text text	author text	created_at timestan
☐	1	1	Can reproduce on both ...	alice@example.com	2026-08-10 21:41
☐	2	1	Looks like a null poin...	dave@example.com	2026-08-10 21:41
☐	3	2	Design mockups are rea...	bob@example.com	2026-08-10 21:41
☐	4	2	Started on the CSS var...	erin@example.com	2026-08-10 21:41
☐	5	3	Confirmed the exported...	carol@example.com	2026-08-10 21:41
☐	6	3	Fixed by appending the...	dave@example.com	2026-08-10 21:41
☐	7	4	Can you send me a scre...	bansari.vadgama@gmail...	2026-08-10 22:13
☐	8	5	I have enabled your ac...	bansari.vadgama@gmail...	2026-08-10 22:15
☐	9	5	Yes, I am now able to ...	bansari.vadgama@gmail...	2026-08-10 22:15
☐	10	6	Sorry, I was using the...	bansari.vadgama@gmail...	2026-08-10 22:16

ticket_messages table, scrolled to show the ticket foreign-key reference column, confirming the ticket_id → tickets.ticket_id relationship:

ticketing-syst...

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Tables

databricks_postgres

Schema

public

Search...

ticket_messages

tickets

137ms

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+

Add record

	cket_id integer	message_text text	author text	created_at timestamp w...	ticket
☐		Can reproduce on both ...	alice@example.com	2026-08-10 21:41:31.11...	ticket
☐		Looks like a null poin...	dave@example.com	2026-08-10 21:41:31.11...	ticket
☐		Design mockups are rea...	bob@example.com	2026-08-10 21:41:31.11...	ticket
☐		Started on the CSS var...	erin@example.com	2026-08-10 21:41:31.11...	ticket
☐		Confirmed the exported...	carol@example.com	2026-08-10 21:41:31.11...	ticket
☐		Fixed by appending the...	dave@example.com	2026-08-10 21:41:31.11...	ticket
☐		Can you send me a scre...	bansari.vadgama@gmail...	2026-08-10 22:13:50.73...	ticket
☐		I have enabled your ac...	bansari.vadgama@gmail...	2026-08-10 22:15:30.62...	ticket
☐		Yes, I am now able to ...	bansari.vadgama@gmail...	2026-08-10 22:15:45.82...	ticket
☐		Sorry, I was using the...	bansari.vadgama@gmail...	2026-08-10 22:16:56.23...	ticket

Note: the tables show 6 tickets / 10 messages total, while the app's stats strip shows 5 tickets / 9 messages — the difference is ticket 6 ("VPN not working") and its one message, which are soft-deleted (deleted_at set) and correctly excluded from the app's active counts while still present in the database.

5. Reflection

What was the most difficult part?

The Lakebase User which I had created did not have permissions for the databricks app . Even though this user was created and had all the right grants, still I was not able to execute SQL Queries from Lakebase so I had to use the native origin user my account for these queries which created issue when deploying the app, so at the end I had manually run the script through command line from my repository to grant these permissions and get the app deployed and running.

How is Lakebase different from storing this data in a traditional analytics table?

Lakebase offers modern features such as support for CDC, DeltaTables supporting modern open table formats like Apache Iceberg which traditional analytics tables does not.

What feature would you add next?

I would have loved to implement the CDC next for getting hands-on only if I had a premium version of Databricks.